

**AN ASSESSMENT OF THE CAUSES AND EFFECTS OF
FAILURE OF BUILDING COMPONENTS IN OWERRI-WEST,
IMO STATE**

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BUILDING COMPONENTS IN OWERRI-WEST, IMO STATE**

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**A RESEARCH PROJECT PRESENTED TO THE
DEPARTMENT OF BUILDING TECHNOLOGY, SCHOOL OF
ENVIRONMENTAL DESIGN AND TECHNOLOGY, FEDERAL
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IN BUILDING TECHNOLOGY.**

**SUPERVISOR
BLDR. OKECHUKWU KELECHI**

NOVEMBER, 2024

CERTIFICATION

This is to certify that this research project on ‘**An Assessment of The Causes and Effects of Failure of Building Components in Owerri-West Imo State**’ is an original work carried out by **Omini Bernard Iwara** with Registration Number **22BH/0105/BT** in the Department of Building Technology, School of Environmental Design and Technology Federal Polytechnic Nekede Owerri, Imo State and it has not been submitted in part or full to any Diploma or degree of this or any other Federal Polytechnic.

APPROVAL

The project title “**An Assessment of The Causes and Effects of Failure of Building Components** in Owerri -West, Imo State prepared by **Omini Bernard Iwara** with Registration Number **22BH/0105/BT** has been approved for the Department of Building Technology, School of Environmental Design and Technology (SEDTE) Federal Polytechnic Nekede Owerri, Imo State.

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DEDICATION

I dedicate this research project to God Almighty who has made it possible to accomplish this success.

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ABSTRACT

The research investigated the causes and effects of failure of building components in Owerri West Area using historical data from 2017 to 2022 and also proffers appropriate solutions. Relevant books, seminar papers, workshop papers, articles, etc. were reviewed so as to examine the general view of individuals that have worked on similar study. Data for the study were obtained through historical data of past building collapse in Nigeria. The data were presented and analyzed using tables, percentage and Chi - square statistical tool. Seventy (70) buildings that collapsed in the country were gathered, upon which the analysis was carried out. The study revealed that poor maintenance culture, design error, poor quality of materials and workmanship, natural phenomenon and excessive loading contributed to about 7%, 13%, 53%, 7%, and 20% respectively of building collapse in Nigeria with most of them being private residential buildings executed by indigenous contractors. The study finally recommended that Standard Organization of Nigeria (SON) should increase their effort in sanitizing building materials in the market. More so, professional bodies in building industry should ensure proper supervision of workmen and efficient checking of materials before incorporation into building works.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The ever-mounting crisis in the housing sector of Nigeria has been in various dimensions, which range and slums, the rising cost of housing rent and the building collapse. The collapse of building has been an issue of great concern to all and sundry in Nigeria and yet these buildings are major contributors to the development of any nation. Therefore, when a building collapses, it fails to render the functions which it should perform to the people and the government. A building is regarded as collapsed if the whole structure or part thereof has fallen down and therefore cannot serve the purpose for which it was intended. When building collapses, it generates deep emotional reactions from every segment of the population constituting great losses to the country in terms of physical, human and material resources Avedun et al., (2013); Nbrri (2020). The incidence of building collapse in Nigerian is reaching an epidemic proportion. These incidences have resulted in the loss of lives and property, which have also infringed permanent disabilities to many. Though building collapse is not peculiar to Nigeria, the trend in the country is becoming quite worrisome and a source of concern to stakeholders. According to Ayedun et al., (2021) the spate and frequency of occurrence have become a major source of concern not only to the governments but to all well-meaning Nigerians and most especially the stakeholders in the building industry as the magnitude of the incidents are becoming very unprecedented. That building collapse incidence is still regularly occurring despite the fact that there has been the increasing diffusion of engineering knowledge over the years have brought to question whether these stakeholders have critically examined the reasons for building failure and the roles they can play or the strategies they can articulate that would help to arrest the incidents. Incidences of building

collapse in Nigeria are posing serious challenges to all the stakeholders in the building industry- building consultants, governments, developers, landlords and users. Others include collapse of Multi-Storey Building in Mokola, Ibadan, Oyo State (1974), building under construction at Benjamin Opara Street, Port Harcourt, Rivers State, (2006) and many others. (Olagunii et al., 2019). On the Night of November, 2012, an uncompleted 3 storey building Collapsed in a water logged area of Owerri during a heavy downpour. Fake cement was used, supervised by a chemist! On May, 15th, a 4 storey building under construction collapsed in Agoama Estate in Omuahia killing undisclosed number of squatters under the floors. Investigation revealed that building regulations permit a maximum of 2 floors in the area. On 5th September, 2013, a 4 storey building under Construction, collapsed at 24 Obanye Street in Onitsha, during a downpour.

In the past few years, incessant building failure has been reported resulting in the loss of lives and properties in Nigeria. However, there has been a dearth of information regarding any findings about the collapse of the building structures. There are frequent media reports of collapsed of buildings in major Nigerian cities like Lagos, Abuja, and Port Harcourt, etc. and other parts of the world. Unfortunately, since independence, the Nigeria government has desperately continued to make concerted effort in the area of quantitative (but not qualitative) supply of mass housing through huge budgetary and policy provisions but, surprisingly, the rate at which existing ones are collapsing calls for an urgent attention. Incidences of building collapse in Nigeria are posing serious challenges to all the stakeholders in the building industry- building consultants, governments, developers, landlords and users. One could imagine what edifices these buildings would have been if only they were constructed accordingly. It has been reported that Nigeria, especially Urban centers has become the “world’s junk-yard” of collapsed

buildings worth billions of naira (Famoroti, 2019). The site of building collapse scattered across the length and breadth of Nigeria is quite alarming that it is unimaginable what effects it will have on the building industry and Nigeria economy as a whole. It is quite unimaginable that a county blessed with so great potentials in its construction industry can experience such magnitude of building collapse.

Fadamiro in 2019 defined building as “an enclosure for spaces designed for specific use, meant to control local climate, distribute services and evacuate waste”. Buildings can be defined as structural entities capable of securing self by transmitting weights to the ground.

1.2 Statement of the Problem

The increasing report of building collapse in Imo State particularly in Owerri Municipal Council has left so many questions unanswered as to what the government is doing to avert the menace.

The unfortunate trending situation may be ascribed to negligence by governments at all levels, professional bodies of the built environment in Nigeria as well as the private Sector and even the consumers of the final product of the industry, all of who are the critical stakeholders in the built environment sector of the economy. Based on the alarming situation of rampant collapse of buildings in Nigeria, this study intends to critically analyze the causes and effects of failure of building components. What have all the professional bodies associated to build environment done to mitigate building collapse and have there been penalties meted out to indicated company organization or client and contractors or even consultants or professionals who may have been indicted.

The researcher believes that the followings could be attributed to the major causes of failure of building components.

- i. Poor Workmanship.
- ii. Improper Specification.
- iii. Uses of Quarks.
- iv. Low Quality of Building Materials.
- v. Overloading and Misuse.
- vi. Inadequate Testing and Quality Control.
- vii. Inadequate Structural Support or Reinforcement.
- viii. Improper Curing of Concrete.
- ix. Incorrect Concrete Mix Ratios.
- x. Improper Load Distribution.

1.3 Aim and Objective of the Study

Aim

The main thrust of this study is to evaluate the causes and effects of failure of building components in Owerri West, Imo State.

Objective

The following are objectives of the study:

- i. To Identify Common Causes of Failure of Building Components within Owerri West Area.
- ii. To ascertain the Effect of Failure of Building Components in Owerri West Area.
- iii. To Assess the Effects of Components Failure on Buildings.

- iv. To Provide Recommendations for Prevention and Management.
- v. To Ascertain Whether Low Quality of Building Materials contribute to Failure of Building Components in Owerri West Area.
- vi. To Ascertain Whether Improper Specification Contribute to Failure of Building Components in Owerri West Area.

1.4 Research Question

- i. What Are the Causes of Failure of Building Components in Owerri West Area?
- ii. What Is the Effect of Failure of Building Components in Owerri West Area?
- iii. Does Low Quality of Building Materials Contribute to The Failure of Building Components in Owerri West Area?
- iv. Does Poor Workmanship Contribute to Failure of Building Components in Owerri West Area?
- v. Do You Think Improper Specification Contribute to Cause of Failure of Building Components in Owerri West Area?

1.5 Research Hypothesis

Hypothesis One

H₀: There Are No Factors That Causes Failure of Building Components in Owerri West Area.

HA: There Are Factors That Causes Failure of Building Components in Owerri West Area.

Hypothesis Two

Ho: Failure of Building Components Do Not Have Any Significant Effect in Owerri West Area.

HA: Failure of Building Components Have Significant Effect in Owerri West Area.

Hypothesis Three

Ho: Low Quality of Building Materials Does Not Contribute to Failure of Building Components in Owerri West Area.

HA: Low Quality of Building Materials Contribute to Failure of Building Components in Owerri West Area.

1.6 Significance of the Study

The following are the significance of this study:

The outcome of this study will educate the general public and the government of the day on the causes, effect and solutions to the cases of building collapse in Nigeria.

This research will also serve as a resource base to other scholars and researchers interested in carrying Out further research in this field subsequently, if applied will go to an extent to provide new explanation to the topic.

1.7 Scope of the Study

This study on residential building collapse in Imo state with careful examination of the causes and the effects with a view of finding a lasting solution to the issue of incessant building collapse in Owerri.

1.8 Definition of Terms

- i. **Building Components**
Building components are the individual parts or elements that come together to form a complete building.
- ii. **Structure:** This is a whole building, complex framework or essential part of a building.
- iii. **Collapse:** This is a state of complete failure.
- iv. **Building Failure:** This is defined as an act of omission or occurrence or performance.
- v. **Building Collapse:** This is a total or a partial/progressive failure of one or more components of a building leading to the inability of the building to perform its principal function of comfort, satisfaction, safety and stability.
- vi. **Building Regulations:** This is a rule prescribe for builder's conduct.
- vii. **Failure:** This is when a construction fails to fulfill its function due to negligence or inexperience of building team.
- viii. **Assessment:** The process of assessing a situation.
- ix. **Causes:** The efficient thing that which make thing to be happened.
- x. **Effect:** A result or product of some cause or agency. I.e. Consequence.
- xi. **Materials:** These are components use to make a building.

CHAPTER TWO

REVIEW OF RELATED LITERATURE

2.1 Building Collapse as a Worldwide Phenomenon

Building collapse is a worldwide phenomenon. Even among the advanced countries, there have been reported cases of collapsed buildings. Quite recently in Riode Janeiro, Argentina, the Associated Press on 30th January, 2012 reported that Bire Bighters pulled out bodies from the rubbles of 3 collapsed buildings. And earlier in September 30, 2010, a 13-Storey building toppled over in China. However, there regularity and causes of occurrence differ from one clime to another. While the causes of building collapse in most other countries are traceable to natural forces (earthquakes, tsunamis, floods, etc.), in Nigeria, its occurrence is mainly due to subjective factors (negligence, Incompetence, greed, etc.) on the part of the professionals and promoters of such building works Okereke et al., (2018) cited in Ezeanyika (2018). The regularity of collapsed buildings across the country is quite alarming and a source of great concern to the built environment professionals and the general public, because of the enormous loss of life and property, each time a building collapsed. Records show that between 1971 to the present, the average number of reported cases of building collapse is not less than 10 annually with not less than 15 lives lost and properties worth millions of Naira wasted.

2.2 Building Collapse

According to the dictionary of Architecture and Construction refers to collapse as mechanical failure. Collapse is a state of complete failure, when the structure has literally given way and most members have caved-in, crumbled or buckled: the building can no longer stand as originally built. Dimunà (2019). A structure is a whole building, complex framework or essential part of a building. According to

Fakere et al., (2018) structure is defined as a body capable of resisting applied loads without any deformation of part relative to one another. The structure of the building is therefore that part of the building construction which gives the construction sufficient strength to withstand the load to which the load is subjected.

Building collapse can simply be defined as a total or a partial/progressive failure of one or more components of a building leading to the inability of the building to perform its principal function of comfort, satisfaction, safety and stability. Olagunju et al., (2021).

A building may collapse when one or more of its essential components fail Fakere et al., (2012). Building failure is defined as an act of omission or occurrence or performance. Failure could also be defined as non-occurrence, non-performance, running short, breaking down, ill-success and, insolvency unsuccessful attempt Ayuba et al., (2017).

Building failure could be of two types namely; cosmetic failure that occurs when something has been added to or subtracted from the building, thus affecting the structures outlooks while structural failures affect both the outlook and structural stability of the building. Ayuba et al., (2017). Hence, failure occurs when a building is not able to perform the function. Ukpata and Sunday, (2020).

Collapse of building either total or partial collapse of some of its components leads to the failure of building to perform its intended function of protection, safety or stability (Olagunyi, et al., 2018)

According to Ikpo, (2017), the degree of building failure can be related to the degree of deviation of the building from its “as built” state which in most cases represents the acceptable standard within the neighborhood, locality, state or country.

2.3 Building Failure

Building Failure is the inability or insufficiency of the building components to perform their expected functions as specified in the design and construction requirements. Building failures can be categorized as physical or structural and performance failures. Physical or structural failure occurs in building due to loss of certain characteristics and strength thereby making such building not strong enough to support the applied load.

Performance failure on the other hand, is the reduction in function of building below an established acceptable limit (Douglas and Ransom 2017).

According to Ede (2018) “Every structural system is designed to meet some needs and be safe to avoid loss of live, property and damage to the environment. In a normal setup, building collapse is not expected within the projected lifespan of structures, but due to the imperfection in the actions of human beings and the existence of so many other external factors that influence the safety of structures, failures do occur.”

Building failure is a common phenomenon all over the world and not peculiar to Nigeria, a number of cases were reported at the international level, but the frequency of occurrence and the consequence devastating effects in the developing countries like Nigeria has become a reoccurring decimal, a worrisome menace nightmare and

an embarrassment. In Asia for instance, Mumbai recorded one of the highest numbers of building collapse between 2008 and 2012. Due to scarcity of land and high rent, constructing new houses was difficult for the middle and low class citizens. They were forced to live in old dilapidated properties which resulted into over 100 building collapses in the city in which 53 people died and 103 others were injured. Lack of maintenance, use of substandard materials, inadequate supervision and non-adherence to building codes were all attributed as the causes of the disaster.

According to Corbett (2015) cited in Babalola (2020) Dhaka also had her share of the menace. The shortage of land and flooding problem facing the country made a factory owner to construct additional floors without reference to building codes and approvals. As a result, a deadliest factory collapse was recorded in 2013 leading to the death of over 547 people.

Structural failure in buildings comes in various forms and degrees of severity ranging from deterioration to decay and a total loss of bearing strength, resulting in sudden breakdown, physical depletion and/or falling apart is termed a collapse. Akinpelu (2002) ascribed environmental changes, natural and manmade hazards, improper presentation and interpretation in the design as the major causes of structural failure in building.

Oloyede et al., (2018) described the causes of building failure as improper design, incompetent contractor, faulty construction methodology, poor Town Planning approval/development monitoring process, non-compliance with specifications/standards by developers/contractors, use of substandard materials and equipment, inadequate supervision or inspection/monitoring, economic pressures,

incompetent conversion, change of use of buildings, aged buildings as well as poor maintenance culture.

Taiwo and Afolami (2017) pointed at inadequacies of empirical rules such poor construction, mistakes by construction supervision and lack of adequate understanding of the new construction technologies and structural intricacies as causes of building failure. Building failure is peculiar to different building situations and when it is not addressed, it results in building collapse.

Adenuga (2019) revealed that the building industry is proliferated with the activities of quacks and inexperienced contractors which have led to collapse of many buildings in Nigeria. He also highlighted poor management and leadership on the part of the Site Engineers and Builders as some of the causes of building failures in Nigeria.

2.4 Building Industry

The building industry is the most complex of all the industries in the economy and the basis of its complexity is founded on the simple fact that, all other industries and sector of the socio-economy depend on it for the environment in which they operate. The building industry is to all practical purpose an all-comers affair Omenihu, Onundi, & Alkali, (2020), It is an industry where all manners of local and foreign materials, professionals and equipment's co-habit in order to achieve quality buildings of high standard. The building industry plays an important and dynamic role in the process of sustainable economic growth and development of any nation due to its size and complexity. It is to be noted that up to one sixth of the total amount allocated to construction projects by Nigeria governments takes the form of building as observed from past budget of the country. Whether a country is just developing like Nigeria or is already developed like Britain, buildings all over the world,

constitute the most valuable assets of mankind (Chinwokwu, (2019). More so, while these buildings provide humanity with a great variety of accommodation in form of residences, mosques, churches, offices, schools, factories, hospitals, stadia, ports, hotels, and so on, it also provides employment for the skilled and unskilled persons.

The building industry plays an important and dynamic role in the process of sustainable economic growth and development of any nation due to its size and complexity. It is to be noted that up to one-sixth of the total amount allocated to construction projects by Nigeria governments takes the form of building as observed from past budget of the country. The purpose of the building industry is to provide suitable accommodation for the whole community, of the quality that can be appreciated by the community, at the cost that the community can afford, within the time required by the community and within the capacity of the building industry. However, it could be deduced that the ultimate goal for any building projects is for such projects to be delivered within the shortest possible time, at the lowest possible cost, within the highest possible quality so as to minimize the problem and the burden of future maintenance and building collapse.

2.5 Identified Causes of Failure of Building Components

Through the author's personal experience and through investigations and media reports, (Olagunju et al., 2021). Identified causes of building collapse in the country can be summarized as follows.

i. Poor Workmanship

Poor workmanship refers to substandard construction practices that compromise the quality and durability of a building. This can result from inadequate skills,

lack of training, or negligence on the part of the workers and supervisors. Poor workmanship can manifest in several ways:

- a. **Improper Material Mixing:** For example, concrete requires precise ratios of cement, sand, water, and aggregates. Incorrect mixing can lead to weak concrete that cannot support structural loads effectively.
- b. **Incorrect Installation Techniques:** Misalignment of structural elements like beams, columns, and walls can lead to load distribution issues, potentially causing cracks and structural failure.
- c. **Neglect of Quality Control:** Failure to inspect materials and construction processes can allow defects to go unnoticed until they cause significant damage. For instance, not checking the consistency of mortar during bricklaying can lead to weak joints.
- d. **Inadequate Curing:** Proper curing is essential to ensure concrete reaches its full strength. Without sufficient curing, concrete may dry too quickly, resulting in shrinkage cracks and reduced durability.

ii. **Improper Specification**

Improper specification is the failure to define the correct materials, methods, and performance standards for a building project. It occurs when project managers, architects, or engineers either overlook or misjudge the requirements of the structure. Examples of improper specification include:

- a. **Choosing Inappropriate Materials:** Each material has unique properties. For instance, using wood in a high-humidity area without proper treatment can lead

to rot. Similarly, using low-quality or incorrect grades of steel or concrete can result in weakened structural integrity.

- b. **Insufficient Load Calculations:** Every component must be able to support anticipated loads, including dead loads (the weight of the building itself) and live loads (people, furniture, equipment). Incorrect specifications might not account for these factors, resulting in structural strain and potential collapse.
- c. **Ignoring Environmental Conditions:** If the building is in an area prone to high winds, earthquakes, or heavy rain, specifications should include materials and designs that can withstand these conditions. For instance, ignoring corrosion protection in coastal areas can lead to rust and weakening of metal components.
- d. **Failure to Follow Building Codes and Standards:** Building codes and standards ensure safety and durability. Incorrect specifications can lead to non-compliance, making buildings unsafe and possibly illegal.

iii. **Use of Quacks**

The term "quacks" refers to unqualified or inexperienced individuals who attempt to take on building projects without the necessary training or expertise. The involvement of quacks often results in the following issues:

- a. **Lack of Technical Knowledge:** Qualified professionals understand structural principles, material properties, and construction techniques, whereas quacks may lack this knowledge. For example, without knowledge of proper load-bearing techniques, they might construct walls or foundations that are too weak.
- b. **Ignoring Safety Standards:** Quacks may not be aware of or adhere to safety standards, leading to unsafe structures that endanger occupants and workers.

c. **Poor Design and Planning:** Proper design and planning are critical to a building's performance. Quacks may lack the skills to interpret blueprints accurately, resulting in mistakes during construction, such as incorrect placements of structural elements.

iv. **Low Quality of Building Materials**

Using low-quality building materials can significantly compromise a structure's durability and safety. These materials often have inconsistent properties, such as lower compressive strength, poor durability, and higher susceptibility to environmental conditions like moisture and temperature changes. For example, substandard concrete can crack under load, and low-quality steel may corrode faster. Ensuring high-quality materials through proper sourcing and testing is essential to prevent premature failures in building components.

v. **Overloading and Misuse**

Overloading occurs when a structure is subjected to loads beyond its design capacity. This can include excessive weight from heavy equipment, storage of materials, or additional floors added without considering the original design limits. Misuse refers to using the structure in ways it was not intended for, like converting a residential building into a heavy storage area. These practices can lead to excessive stress on structural components, resulting in cracks, deformation, and even collapse.

vi. **Inadequate Testing and Quality Control**

Quality control and testing ensure that materials meet the required standards and specifications before and during construction. Inadequate testing, such as not checking concrete's compressive strength or the tensile strength of steel, can lead to unforeseen weaknesses in the structure. Quality control includes regular

inspections, material testing, and verification of construction practices. Without these measures, faults go undetected, potentially leading to structural issues and unsafe conditions.

vii. **Inadequate Structural Support or Reinforcement**

Proper structural support, such as beams, columns, and reinforcements, is essential to maintain the integrity of a building. Inadequate reinforcement can lead to weak points within the structure, reducing its ability to withstand applied loads. For example, insufficient rebar in concrete columns can reduce their load-bearing capacity, leading to cracking or buckling. Reinforcement must be correctly designed and installed to handle the expected loads and environmental stressors.

viii. **Improper Curing of Concrete**

Curing is the process that ensures concrete reaches its full strength by retaining moisture over a specific period. Improper curing, such as premature drying or exposure to extreme temperatures, can result in weak, brittle concrete. This can lead to cracking and reduced load-bearing capacity. Proper curing methods, such as keeping the concrete covered or spraying it with water, allow it to gain the necessary strength for long-term durability.

ix. **Incorrect Concrete Mix Ratios**

Concrete's strength and durability depend heavily on the correct proportions of cement, sand, aggregate, and water. An incorrect mix ratio can result in either too much or too little water, affecting the concrete's compressive strength. For example, excess water weakens the mix, making it prone to cracks and structural failure, while insufficient water can make it challenging to work with, leading to incomplete bonding between materials.

x. **Improper Load Distribution**

Improper load distribution refers to the uneven transfer of weight through structural components. This imbalance can be due to improper design, poor construction practices, or changes in usage. Uneven distribution causes excessive stress on certain areas, resulting in cracks, deflection, or even collapse. Proper structural analysis and design are necessary to ensure loads are distributed evenly, with adequate supports and reinforcements in place.

xi. **Defective Design**

Defective Architectural and Engineering drawing may result if architects fail to do or insist on the carrying out of feasibility studies, soil and site investigation which are the bases for design of adequate architectural and structural drawings. Others include poor design details, low quality materials and works specifications. Engineers (structure and civil) may contribute if they fail to insist on carrying out essential soil test, foundation design, errors, omissions and inaccurate data from professionals may lead to problems if not detected on time.

xii. **Defective Construction**

This arises when contractors fail to carry out the works in accordance with architects and engineers' specifications. They do this in order to maximize profit. Sometimes specified materials are substituted for substandard ones. Others areas of Concern include poor concrete mixes, premature removal of formworks and general poor workmanship.

xiii. **Absence of Building or Planning Permit**

It is illegal to commence construction works without approved drawings from the approving authorities. The 3 tiers of government- the Commission (for Federal Lands), the Board (for State lands), and the Authority (for Local Governments

lands), are vested with the duties of granting approval to prospective developers. Sometimes defective drawings are used for construction without approval from the approving authorities. Some are done out of ignorance. Others are done where the operation of the authorities are ineffective.

xiv. **Corruption**

Sometimes drawings are not read by officers of the approving authority to detect defects. They sometimes engage in corrupt practices by granting illegal approvals.

xv. **Non-Adherence to Approved Building Plans**

This comes in form of illegal alteration to approved drawings. Sometimes, a building originally specified to undergo in-situ concreting is changed to pre-cast methods because the expatriate contractor tends to prefabricate the components overseas and ship to Nigeria. This practice if not properly controlled could spell danger years after the buildings are in use.

xvi. **Absence of proper site and soil investigation**

The avoidance of this to determine suitability of the terrain soil's bearing capacity, which influences foundation types spells danger. Engagement of inexperienced personnel to take charge of construction works: Sometimes firms resort to use unqualified staff to act as principals on construction sites in order to save cost. The consequence of this practice is that the unqualified staff may not be competent enough to detect fraudulent practices of smart contractors. This may lead to covering up of shoddy or defective works.

xvii. **Engagement of ill-equipped, incompetent contractors**

In Nigeria, it is common that most contracts are first awarded to businessmen who front for politicians. The practice is that the businessman gets the contract and

sells it to incompetent contractors known to them without following the normal contract procedures and without investigating the competency of the contractor. The result is shoddy performance which can lead to building collapse,

xviii. **Lack of proper supervision, inspection and monitoring of construction works**

Building professionals both in practice and in government agencies more often are guilty of this practice either due to negligence or they are not paid to do so.

xix. **Illegal conversion, alteration, and additions to existing structures.**

Imposing additional floors beyond original design provision is a common practice in Nigeria. For examples, a storey building changing to 2 storey thereby imposing more loads on the suspended floor with resultant load on foundation. Creating additional rooms on suspended floors, changing the use of building, for example, converting residential to mini-factory whereby heavy duty equipment are placed on suspended floors. Also creating vibrations on suspended floors by breaking of slabs, beams, or exposing old foundation to flood and erosion may lead to structural failure.

xx. **Undue Interference of client on building works:**

Sometimes the client makes serious changes and variations at advanced stage of construction with the contractor without seeking building consultant's advice.

xxi. **Foundation failures:**

A building structure can collapse if founded on poor sub-soil, or if the building is not uniformly loaded or if suitable foundation was not specified according to soil nature or due to soil erosion or earth movement under the foundation.

xxii. **Fire Outbreak:**

Most materials used in building construction and finishes are flammable (Olagunju, 2021). These materials for example, gloss paints encourage fire spread. Fire weakens structural members such as reinforcement bars and steel trusses. These materials fail in the process of providing supports to components and main structure and in the event of fire, may lead to total or partial collapse of the building.

xxiii. **Natural Occurrences**

In Nigeria, the most common natural events are heavy storm, high wind, flood, thunder, and lightening. Many reported cases of building collapse in Nigeria caused by natural disasters have been reported by Arayela and Adam (2018). An example is a 3 Storey residential building, at Iju-Ishaga, Lagos (September, 1999)

2.6 Building Components

Building components are the individual parts or elements that make up a structure. They can be categorized into various systems based on their functions and roles in construction. Here are some key building components.

i. **Foundations**

Purpose: Foundations are the base of a building, designed to evenly distribute the structure's weight onto the soil and provide stability.

Types: Common types include shallow foundations (e.g., spread footings) for lighter structures and deep foundations (e.g., piles and drilled shafts) for larger buildings or areas with weak soil.

Failure Risks: Poor soil testing, inadequate depth, or substandard materials can lead to settlement, cracking, or complete foundation failure.

ii. **Columns**

Purpose: Columns are vertical members that support loads transferred from beams and slabs to the foundation.

Types: They can be load-bearing or non-load-bearing and are often categorized based on their materials (e.g., concrete, steel, wood).

Failure Risks: Issues such as buckling, corrosion, or inadequate load capacity can lead to instability, making columns a critical component that requires proper reinforcement.

iii. **Beams**

Purpose: Beams are horizontal members that carry loads from slabs or roofs and transfer them to columns or walls.

iv. **Types:** Common types include reinforced concrete beams, steel beams, and timber beams. They are categorized by shape (e.g., I-beam, T-beam) or material.

v. **Failure Risks:** Improper design, overloading, or material fatigue can cause beams to crack or deflect, impacting the structure's overall stability.

vi. **Slabs**

Purpose: Slabs form floors and ceilings and provide horizontal surfaces for occupancy and separation between building levels.

Types: They include solid slabs and hollow-core slabs made from materials like reinforced concrete or steel.

Failure Risks: Poorly constructed slabs can crack or sag due to excessive loads, shrinkage, or substandard materials, leading to safety hazards.

vii. **Walls:**

Purpose: Walls provide vertical enclosure, privacy, and support. They can be load-bearing or non-load-bearing.

Types: Walls are categorized by function (e.g., exterior walls, partition walls) and materials (e.g., concrete, brick and drywall).

Failure Risks: Cracking, moisture infiltration, and structural weaknesses can occur due to poor material choice, exposure to weather, or inadequate support.

viii. **Roofs**

Purpose: Roofs protect the building from weather, provide insulation, and sometimes support additional load (e.g., green roofs).

Types: Flat, gabled, hipped, and shed roofs, with materials ranging from asphalt shingles and metal to concrete.

Failure Risks: Leakages, sagging, and collapse may result from poor design, inadequate drainage, or weather-related wear and tear.

ix. **Doors and Windows**

Purpose: These components provide access, ventilation, natural light, and security.

Types: Doors can be interior or exterior, sliding or hinged, while windows vary in type (e.g., fixed, sliding) and material (e.g., wood, aluminum, UPVC).

Failure Risks: Malfunction, breakage, or weathering of doors and windows can lead to loss of energy efficiency, security, and comfort.

x. **Stairs**

Purpose: Stairs provide access between different building levels and serve as emergency exits.

Types: Types include straight, spiral, and winding stairs, made from materials like concrete, steel, and wood.

Failure Risks: Poor design or inadequate support can cause stairs to sag or collapse, which is hazardous for occupants.

2.7 Effects of Failure of Building Components

Akinpelu, (2019) identified the following effects on the failure of building components.

- i. Loss of life, property and huge sum of capital. About 217 people lost their lives between 2014-2019 from Newspaper survey conducted by Arayela and Adam (2020)
- ii. Loss of reputation and integrity leading to psychological trauma
- iii. Loss of new commissions and contracts
- iv. Withdrawal of practicing licenses

- v. **Loss of materials and capital investments:** Components and materials are damaged beyond re-use. Capital investments are not recoverable, leading to bankruptcy and high economic implications to the nation's economy.
- vi. **Structural Integrity:** Compromised components, such as beams or columns can weaken the building's structural integrity, leading to partial or total collapse.
- vii. **Safety Hazards:** Failures can create dangerous conditions for occupants, such as exposure to falling debris fire hazards from faulty electrical systems or water damage from broken plumbing.
- viii. **Aesthetic Damage:** Components failures can cause visible damage to the building's appearance, such as cracks in walls or peeling paints affecting its overall aesthetic.
- ix. **Compromise security:** Compromised failures in doors, window or locks can undermine building security, increasing their risk of unauthorized access or burglary.
- x. **Aesthetic Degradation:** Persistence components failures can lead to visible damages, such as stained walls or sagging ceiling which diminishes the building overall appearance.
- xi. **Compromise Accessibility:** Failure in elevators, ramps, or doorways can hinder access for individual with disabilities, impacting the building's compliance with accessibility standards and reducing inclusivity.

2.8 Remedies to Causes of Building Collapse

Based on the causes of building collapse identified in this study, it may be argued (and rightly) that the remedies are direct opposite of the causes. In the light of this. Some of the remedies to the causes of building collapse already identified are hereby highlighted as,

- i. Discouraging the use of quacks for construction Works, encouraging the use of reputable contractors for the building works,
- ii. Establishing "Developers' Aid Unit" in government professional offices at all levels.
- iii. Prompt payment of certified Jobs by government and individuals.
- iv. Planned timely maintenance of property, compulsory implementation of National Housing Policy, discouraging property alterations and adherence to property use without obtaining appropriate approvals.
- v. Encouraging soil geotechnical investigation, establishment of material testing laboratories across the country for easy reach, ensuring the compulsory use of professionals for project execution, encouraging the publication of results of enquiries on collapsed buildings.
- vi. Ensuring that adequate penalties are meted out to defaulting professionals and or professional bodies.
- vii. In awarding contracts, governments should consider the employer-employee relationship and the law which assumes that anyone employed to execute construction work is trained and skillful enough to execute the project. It then behooves on anyone employing a contractor to ensure that the contractor has

the expertise to carry out the work. Such knowledge is not the privilege of quarks.

- viii. Quacks must not under any pretext be employed by any government or individual to undertake construction works.
- ix. Prompt payment by the government at the local, state or federal level for certified jobs are important.
- x. To enhance contractor's cash flow which will further encourage speedy and timely completion and delivery of projects.
- xi. Governments must evolve an effective maintenance culture in order to prolong the values and life span of these construction products.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Research Design

Research design relates to the general approach adopted in executing the study.

Descriptive survey research design was adopted for the study.

According to Adeniji (2016), descriptive survey is meant to examine the situation as it currently obtainable, and no variable was manipulated. The method enables the researcher to collect data and information from the respondents directly in the field.

3.2 Area of the Study

The study was carried out in Owerri West is a Local Government Area of Imo State, Nigeria. Its headquarters are in the town of Umuguma. Owerri West was carved out of the former Owerri Local Government Area in 1996. A very large portion of the local government constitute the capital city of Imo State, Nigeria. Many business and investment opportunities exist in the council because of the vantage position, the council occupies as the seat of Government and therefore the epicenter of all economic, social, religious activities in the state. In the past Owerri had barely as low as 5% record of building collapse but in the recent years the Occurrence of building collapse in the area has risen and its implication on human lives and properties have given rise to many questions.

3.3 Population of the Study

Population refers to all individuals under study. The population of this study therefore Consist of the construction professionals selected from the list provided

through membership list solicited from the secretariat of their various State Chapter professional bodies. The population is made up of 150 professionals.

3.4 Sample Size and Sampling Technique(s)

Sample constitutes the -part of the population drawn to represent the entire population in the research. The sample size was derived using Taro Yamen's Formula at 5% level of Significance.

Where $n = \frac{N}{1 + N(e)^2}$

$$1 + N(e)^2$$

n = sample size (to be determined)

N = population size

E = level of significance

$N = ?$

$N = 150$

Error = 5%

Thus $n = \frac{150}{1 + 150(0.52)^2}$

$$= \frac{150}{1 + 150(0.0025)}$$

$$= \frac{150}{1 + 0.375}$$

$$= \frac{150}{1.375}$$

$$= 109$$

Sampling Techniques

This has to do with the methods adopted in obtaining my sample size from the entire population of the study. The simple random sampling techniques was adopted and it was used to enable every individual have equal chance of being selected

3.5 Instrument for Data Collection

The questionnaires were structured in open and close ended questions to give the respondents, the opportunities to express their options and views freely and easily section A contains the respondent's personal information while section B contains questions to address the research problems.

3.6 Validity of the Instrument

The research work was validated by the project supervisor by making some correction and amendment. Before it came out, all the criticisms were dully taken into consideration before it was distributed to the respondents. In abide to ensure a more valid work, the researcher embarked on pilot test. This involves the administering of questionnaire yes to other banks in Owerri outside which will not be used as the study similar answers was gotten.

Also, interview was also scheduled on the day of collection of the questionnaires to validate the information in the questionnaire. This is the extent of accuracy of the instrument used in the study. The instrument used in the study is questionnaire. The questionnaire was structures in good manner and it was thoroughly read and approved by the project supervisor who is an expert on this field. The reason for this is for correction of error and to check if the questionnaire answers the research questions and the objectives of the study. The corrections were integrated, which led to the production of the final draft.

3.7 Reliability of the Instrument

Reliability ensures that instrument gives consistent result, often using it within some intervals. To establish reliability of the instruments, a result method is used. The instrument was administered on respondents and their responses were noted. After two weeks, instrument was administered and sources corrected with first results.

3.8 Method of Data Collection

The collection of information for this work was based on primary and secondary data. The primary data includes the following:

Questionnaire

This is a set of printed or written questions with a choice of answer device for the purpose of a survey.

Personal Interview

This involves face-to-face interaction where questions are asked personally to respondent(s)

Observation

This involves gathering data by watching people or events by the researcher. The secondary data includes materials from internet, journals and other unpublished materials.

3.9 Method of Data Analysis

To facilitate an easy understanding of data analysis, the researcher used frequencies and simple percentage method. A percentage rank indicates the value

that fails below a given Score and it seeks to show the relative relationship value and data. The percentage formula used was:

$$\text{Simple percentage} = \frac{F}{N} \times 100$$

Where F = No of respondent

N = Total sum of all respondent

The researcher also used chi-square (χ^2) test statistics to test the hypothesis

Chi-square formula used was:

$$\chi^2 = \frac{(O_i - E_i)^2}{E_i}$$

Where χ^2 is calculated chi-square value

O_i is observed frequency

E_i is expected frequency

Σ = is summation

Generally, the observed frequency refers to the number of times a set of events occur. Thus, the expected frequency (E_i) is calculated as shown below:

$$\text{Expected Frequency } (E_i) = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Where R = Total number of rows

C = Total number of columns

N = Summation of rows and columns

The degree of freedom here is given by:

$$Df = (R-1)(C-1)$$

The Level of significance for calculating the chi-square is 0.05

Decision Rule

If the calculated value of χ^2 is greater than the critical value, we reject the null hypothesis (H_0) and accept the alternative hypothesis (H_1) but if the calculated value of χ^2 is less than the table value (critical value) we accept the null hypothesis (H_0) and reject the alternative hypothesis (H_1)

CHAPTER FOUR

DATA PRESENTATION AND ANALYSIS

4.1 Presentation of Data

Analysis of Questionnaire Distributed

Administered	%	Returned	%	Not returned
109	100	105	96	4

Total of one hundred and five (105) questionnaire were distributed out of which one hundred and five were duly completed and returned while four (4) were not returned.

4.2 Analysis of Data

Table 4.2.1: Gender of the respondents

Gender	No of respondents	%
Male	90	86
Female	15	14
Total	105	100

Source: Survey, 2024

Table 4.2.1 Reveal that 90 respondents (86%) were male while 15 respondents (14%) were female.

Table 4.2.2: Age of the Respondents

Age Range	No of respondents	%
25 – 35 years	5	5
36 – 50 years	10	9
Above 50 years	90	86
Total	105	100

Source: Survey, 2024

Table 4.2.2 indicates that 5 respondents (5%) are between the age of 25 – 35 years, 10 respondents (9%) are between 36 – 50 years while 90 respondents (86%) are above 50 years.

Table 4.2.3: Marital Status

Option	No of respondents	%
Single	25	24
Married	80	76
Total	105	100

Source: Survey, 2024

Table 4.2.3 reveals that 25 respondents (24%) are single while 80 respondents (76%) are married couples.

Table 4.2.4 Working Experience

Option	No of Respondents	%
5 – 10 Years	5	4
11 – 20 years	30	29
Above 20 years	70	67
Total	105	100

Source: Survey, 2024.

Table 4.2.4 shows that 5 respondents (4%) have work experience of 5 - 10 years, 30 respondents (29%) have worked for between 11 – 20 years. While 70, respondents (67%) have worked for more than 20 years.

Table 4.2.5: Academic Qualification

Option	No of Respondents	%
ND/NCE	10	9
HND/BSC	90	86
MSc	5	5
Total	105	100

Source: Survey, 2024.

Table 4.2.5 shows that 10 respondents (9%) are ND/NCE, 90 respondents (86%) are HND/BSc holders while 5 respondents (5%) have M.Sc.

Section B

Analysis on the response to question 1: what are the causes of failure of building components in Owerri West Area?

Table 4.2.5: Poor Workmanship

Option	No of Respondents	%
Poor Workmanship	12	12
Improper Specification	8	8
Uses of Quacks	15	14
All of the Above	70	67
Total	105	100

Source: Survey, 2024.

Table 4.2.6 reveals that 12 respondents (11%) believe that poor workmanship is the cause of failure building components in Owerri West Area, 8 respondents (8%) believe that improper specification is the cause of failure building components in Owerri West Area, 15 respondents (14%) believe that use of quarks is the cause of failure of building components. while 70 respondents (67%) believe is all of above.

Table 4.2.6 Analysis on the response to Question 2: what is the effect of failure of building components in Owerri West Area?

Table 4.2.6: Loss of life, Property and Capital Sum

Option	No of Respondents	%
Loss of life, property and capital sum	25	24
Loss of reputation and integrity	10	9
Loss of material and capital investment	5	5
All of the Above	65	62
Total	105	100

Source: Survey, 2024.

Table 4.2.7 indicates that 25 respondents (24%) says that the effect of failure of building components is loss of life, property and capital sum, 10 respondents (9%) says that the effect of failure of building components is loss of reputation and integrity, 5 respondents (5%) says that the effect of failure of building components is loss of material and capital investment. while 65 respondents (62%) say all of the Above.

Analysis on the response to question 3. Does low quality of building materials contribute to failure of building components in Owerri West Area?

Table 4.2.7: Low Quality of Building Materials

Option	No of respondents	%
Yes	90	86
No	15	14
Total	105	100

Source: Survey, 2024.

Table 4.2.7 reveals that 90 respondent (86%) agree that low quality of building materials contribute to failure of building components in Owerri West Area. while 15 respondents (14%) disagree.

Analysis on the response to question 4: Does poor workmanship cause failure of building components in Owerri West Area?

Table 4.2.8: Failure of Building Components

Option	No of Respondents	%
Yes	65	62
No	40	38
Total	105	100

Source: Survey, 2024.

Table 4.2.8 reveals that 65 respondents (62%) believe that poor workmanship cause failure of building components. While 40 respondents (38%) disagree.

Analysis on the response to Question 5: Do you think improper specification cause failure of building components in Owerri West Area?

Table 4.2.9: Improper Specification Cause Failure of Building Components

Option	No of Respondents	%
Yes	75	71
No	30	29
Total	105	100

Source: Survey, 2024.

Table 4.2.9 shows that 75 respondents (71%) agree that improper specification cause failure of building components in Owerri West Area. while 30 respondents (29%) disagree.

Analysis on the response to Question 6: Does use quarks cause failure of building components in Owerri West Area?

Table 4.2.10: Use of Quarks Cause Failure of Building Components

Option	No of Respondents	%
Yes	90	86
No	15	14
Total	100	100

Source: Survey, 2024.

Table 4.2.10 indicates that 90 respondents (86%) agree that use of quarks cause failure of building components in Owerri West Area. while 15 respondents (14%) disagree.

4.3 Test of Hypothesis

Hypothesis One

Ho: There are no factors that cause failure of building components in Owerri West Area.

HA: There are factors that cause failure of building components in Owerri West Area.

Contingency Table 1A

Option	Male	Female	Total
Yes	80	20	100
No	-	5	5
Total	80	25	105

Calculation of Expected Frequency

RT X CT

Yes

$$\text{Male} = \frac{80 \times 100}{105} = 76.19$$

$$\text{Female} = \frac{25 \times 100}{105} = 23.81$$

No.

$$\text{Male} = \frac{80 \times 5}{105} = 3.81$$

$$\text{Female} = \frac{25 \times 5}{105} = 1.19$$

Calculation of chi-square

$$\frac{(O_i - E_i)}{E_i}$$

O _i	E _i	(O _i -E _i)	(O _i -E _i) ²	(O _i – E _i) ² /E _i
80	76.19	3.81	14.5161	0.191
20	23.81	-3.81	14.6161	0.609
0	3.81	-3.81	14.5161	3.81
5	1.91	3.81	14.5161	12.198
				X ² = 16.808

$$DF = (R - 1) (C - 1)$$

$$(2 - 1) (2 - 1)$$

$$1 \quad 1$$

$$= 1$$

Tested at 0.05% level of significance – 3.841

Decision Rule

Since the chi-square calculated 16.808 is greater than the critical value 3.841, we reject null hypothesis (H₀) and accept the alternate hypothesis (H_A) stating that there are factors that cause failure of building components in Owerri West Area.

Hypothesis Two

H₀: Failure of building components do not have any significant effect in Owerri West Area.

H_A: Failure of building components have any significant effect in Owerri West Area.

Contingency Table 4A

Option		Male	Female	Total
Yes		75	20	95
No		5	5	10
Total		80	25	105

Calculation of expected frequency

RT x CT

GT

Yes

$$\text{Male} = \frac{80 \times 95}{105} = 72.38$$

$$\text{Female} = \frac{25 \times 95}{105} = 22.62$$

No.

$$\text{Male} = \frac{25 \times 10}{105} = 7.62$$

$$\text{Female} = \frac{25 \times 10}{105} = 2.38$$

Calculation of chi-square

$(O_i - E_i)^2$

E_i

Contingency Table 2B

O _i	E _i	(O _i -E _i)	(O _i -E _i) ²	(O _i – E _i) ² /E _i
75	72.38	2.62	6.8644	0.095
20	22.62	2.62	6.8644	0.303
5	7.62	2.62	6.8644	0.901
5	2.38	2.62	6.8644	2.884
				X ² = 4.183

$$DF = (R - 1) (C - 1)$$

$$(2 - 1) (2 - 1)$$

$$1 \times 1$$

$$= 1$$

Tested at 0.05% level of significance = 3.841

Decision Rule

Since the chi-square calculated 4.183 is greater than the critical value 3.841, we reject null hypothesis (H₀) and accept the alternate hypothesis (H_A) stating that failure of building components has significant effect in Owerri West Area.

Hypothesis Three

H₀: Low quality of building materials does not contribute to failure of building components in Owerri West Area.

H_A: Low quality of building materials contributes to failure of building components in Owerri West Area.

Contingency Table 3A

Option	Male	Female	Total
Yes	45	22	67
No	35	3	38
Total	80	25	105

Calculation of expected frequency

RT X CT

GT

Yes

$$\text{Male} = \frac{80 \times 67}{105} = 51.05$$

$$\text{Female} = \frac{25 \times 67}{105} = 15.95$$

No

$$\text{Male} = \frac{80 \times 35}{105} = 28.95$$

$$\text{Female} = \frac{25 \times 38}{105} = 9.05$$

Calculation of chi-square

$(O_i - E_i)^2$

E_i

Contingency Table 3B

O _i	E _i	(O _i – E _i)	(O _i – E _i) ²	(O _i – E _i) ² /E _i
45	51.05	-6.05	36.6025	0.717
22	15.95	6.05	36.6025	2.295
35	28.95	6.05	36.6025	1.264
3	9.05	-6.05	36.6025	4.044
				X ² = 8.32

$$DF (R - 1) (C - 1)$$

$$(2 - 1) (2 - 1)$$

$$1 \quad 1$$

$$= 1$$

Test at 0.05% level of significance = 3.841

Decision Rule

Since the chi-square calculated 8.32 is greater than the critical value 3.841, we reject null hypothesis (H₀) and accept the alternate hypothesis (H_A) stating that Low quality of building materials contribute to failure of building components in Owerri West Area.

4.4 Discussion of Findings

Table 4.2.1 reveals that 90 respondents (86%) were male while 15 respondents (14%) were female.

Table 4.2.2 indicates that 5 respondents (5%) are below the age of 30 years, 10 respondents (9%) are between 31 – 40 years while 90 respondents (86%) are above 40 years.

Table 4.2.3 reveals that 25 respondents (24%) are single while 80 respondents (76%) are married couples.

Table 4.2.4 shows that 10 respondents (9%) are ND/NCE holders, 90 respondents (86%) are HND/BSC holders while 5 respondents (5%) have M.Sc.

Table 4.2.5 reveals that 12 respondents (11%) believe that poor workmanship is the cause of failure building components in Owerri West Area, 8 respondents (8%) believe that improper specification is the cause of failure building components in Owerri West Area, 15 respondents (14%) believe that use of quarks is the cause of failure of building components. while 70 respondents (67%) believe is all of above

Table 4.2.8 reveals that 65 respondents (62%) believe that poor workmanship cause failure of building components. While 40 respondents (38%) disagree.

Table 4.2.9 shows that 75 respondents (71%) agree that improper specification cause failure of building components in Owerri West Area. while 30 respondents (29%) disagree.

2024.

Table 4.2.10 indicates that 90 respondents (86%) agree that use of quarks cause failure of building components in Owerri West Area. while 15 respondents (14%) disagree.

The hypotheses testing established that there are factors that cause failure of building components and failure of building components have significant effect in Owerri West Area.

The hypothesis further established that low quality of building materials contributes to failure of building components in Owerri West Area.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

The analysis of the study revealed the following major findings:

- i. There Are Factors Responsible for Failure of Building Components in Owerri West Area.
- ii. Failure Of Building Components Have Significant Positive Effect in Owerri West Area.
- iii. The Low Quality of Building Materials Significantly Contribute to Failure of Building Components in Owerri West Area.

5.2 Conclusion

The problem of building collapse in Nigeria will continue reverberate in this country unless all critical stakeholders work together with a sincerity of purpose to stem the tide. It has been stated that the building collapse phenomena is largely traceable to unethical practices of some professional stakeholders and developers, including government officials. It must be mentioned that existing regulations and codes are enough to deal with the situation, but these are often adhered to in breach than in observance. The study believes that some government officials contribute to this mess as they collect gratifications and award contracts to incompetent contractors and even issue approvals without actually scrutinizing the projects. Such compromises have caused enormous cost and liabilities to the society who often becomes victims of the collapsed building. Additionally, it is further believed that when buildings collapses, clients do not get value for their investments. The

construction industry will be better able to continue to provide the quality services on which society increasingly depends.

5.3 Limitations of Study

Financial constraint- Insufficient fund tends to impede the efficiency of the researcher in sourcing for the relevant materials, literature or information and in the process of data collection (internet, questionnaire and interview).

Time constraint- The researcher will simultaneously engage in this study with other academic work. This consequently will cut down on the time devoted for the research work.

5.4 Recommendations

- i. Professionals should endeavor to attend conferences /workshops and seminars both local and overseas so as not to put themselves out of rhythm.
- ii. It further recommends the formation of a Federal Building Safety Corps to address all problems of construction in buildings.
- iii. Contractors are charged to use good quality materials as must have been specified in the course of construction.
- iv. At the same time, Government and developers should also deploy the services of construction professionals to every project as stipulated by existing statutes as this will greatly avert the incidences of building collapse in Nigeria.

5.5 Suggestions for Further Studies

- i. Failure of building components in Owerri, Imo State, problems and prospect
- ii. Remedies to the causes of failure of building components in Owerri.
- iii. Effect of building collapse on economic growth in Nigeria.

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APPENDIX

Department of Building Technology,
School of Environmental Design and Technology,
Federal Polytechnic Nekede,
P.M.B 1306 Owerri,
Imo State.

5th October, 2024.

Dear Sir/Madam,

RESEARCH QUESTIONNAIRE

I am HND student of the above mentioned institution, I am conducting a research on **“An Assessment of The Causes and Effects of Failure of Building Components in Owerri-West Imo State** as a requirements for the Award of Higher National Diploma (HND) in the above department.

You are kindly requested to render assistance by responding fully and frankly to, the questions. The received responses will be treated with strict confidentiality and will be used for the research work only.

Yours faithfully,

Omini Bernard Iwara
22/BH/0105/BT
Researcher

Questionnaire Interview Schedule for Computerization of Banks

Introduction: Fill in the gap the gal and tick in () the box where appropriate

Name of Respondents.....

Sex:

i. Male ☐

ii. Female ☐

Position:.....

Age:

i. 25 - 35 ☐

ii. 36 - 50 years ☐

iii. Above 50 years ☐

Working experience:

i. 5 – 10 years ☐

ii. 11 - 20 years ago ☐

iii. Above 20 years ☐

Qualifications:

i. ND/NCE ☐

i. HND/BSc ☐

ii. MSc/PHD ☐

SECTION B

1. What are the causes of failure of building components in Owerri West Area?

- i. Poor workmanship ☐
- ii. Improper specification ☐
- iii. Use of quarks ☐
- iv. Low Quality of Building Materials ☐
- v. Overloading and Misuse ☐
- vi. Inadequate Testing and Quality Control ☐
- vii. Inadequate Structural Support or Reinforcement ☐
- viii. Improper Curing of Concrete ☐
- ix. Incorrect Concrete Mix Ratios ☐
- x. Improper Load Distribution ☐

2. What Is the Effect of Failure of Building Components in Owerri West Area?

- i. Loss of life, property and capital sum
- ii. Loss of reputation and integrity
- iii. Withdrawal of Practicing License
- iv. Loss of materials and capital investments

3. Does Low Quality of Building Material Contribute to The Failure of Building Components in Owerri West Area?

- i. Yes ☐
- ii. No ☐
- iii. Maybe ☐

4. Does Poor Workmanship Cause Failure of Building Components in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐

5. Do You Think Improper Specification Cause Failure of Building Component in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐

6. Does Use of Quarks Cause Failure of Building Components in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐

7. Do You Think Overloading and Misuse Cause Failure of Building Component in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐

8. Do You Think Inadequate Testing and Quality Control Cause Failure of Building Component in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐

9. Does Inadequate Structural Support or Reinforcement Cause Failure of Building Components in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐

10. Do You Think Improper Curing of Concrete Cause Failure of Building Component in Owerri West Area?

- i. No ☐
- ii. Yes ☐
- iii. Maybe ☐